Project 3: Customer Loyalty

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Scenario

Our company, **TableGram**, has strong acquisition numbers but struggles to retain customers long term. Despite frequent usage, many customers, whom we dub **"Promotional Nomads,"** exhibit loyalty challenges**,** such as switching between apps to chase discounts, discontinuing usage after promotions end, expressing dissatisfaction with delivery times or food quality. We need to understand the key **drivers** and **barriers** to loyalty and develop **loyalty-enhancing strategies**.

Outline of Tasks

1. We designed a **survey** and created a **C-suite presentation** outlining its (simulated) insights, tied to specific business strategies.
2. Based on the simulated findings of the survey, we proposed an **A/B test** to evaluate an **intervention** aimed at improving customer loyalty, and simulated the results.
3. Following the survey and A/B test, we developed a **quantitative analysis plan** outlining the information yielded by each step and the **actionable insights** generated by this analysis.

Survey

To capture the "Promotional Nomad," the survey needs to be punchy, mobile-optimized, and psychologically designed to uncover **why** they keep the competition's apps on their home screen.

Here is the functional survey design, structured to move a user from "Price Chaser" to "Brand Advocate."

Survey Design

Section 1: The Pulse

Goal: Determine if loyalty is blocked by price, service, or lack of habit.

1. When you decide to order food, what is the FIRST thing you do?
- [ ] Open TableGram immediately.
- [ ] Open whichever app has a notification/discount code.
- [ ] Check 2–3 different apps to compare total prices.
- [ ] Search Google/Maps for a specific restaurant first.

2. Rank the following in order of importance for your "Default" app (1 = Most Important):
- [ ] Lowest total cost (Fees + Food).
- [ ] Speed of delivery.
- [ ] Accuracy of estimated arrival time (Reliability).
- [ ] Selection of exclusive restaurants.
- [ ] Quality of customer support if something goes wrong.

Section 2: The Competitive Gap

Goal: Identify the specific "Hook" our competitors have.

3. Which of these statements best describes your relationship with other delivery apps?
- [ ] I have a paid subscription (e.g., DashPass, Uber One) elsewhere.
- [ ] I only use them when they send me a "40% off" notification.
- [ ] I prefer their user interface/tracking map more than TableGram.
- [ ] I don't use other apps; I'm loyal to TableGram.

4. What is the #1 reason you choose a competitor over us?
- [ ] Cheaper delivery fees.
- [ ] Better rewards/points program.
- [ ] They have a specific restaurant you don't.
- [ ] Their app is faster/easier to use.

Section 3: The Quality Check

Goal: Determine whether the product is "broken" or just "boring."

5. In your last 3 orders with TableGram, how often did the following occur?
(Scale: Never | Once | Frequently)

- Food arrived cold or messy.
- The delivery was later than the original ETA.
- Hidden fees appeared at the final checkout screen.
- Items were missing from the order.

Section 4: The Future

Goal: Test which loyalty features actually move the needle for a Nomad.

6. Which of these "Member Perks" would make you stop checking other apps? (Select your Top 2)

- [ ] **The "No-Fee" Pass:** A monthly subscription that pays for itself in 2 orders.
- [ ] **Priority Routing:** Your food is always the first drop-off (no "doubling up").
- [ ] **The Streak Bonus:** Get 10% back in credit for every 3 orders made in a month.
- [ ] **VIP Support:** Immediate human chat for any order issue.
- [ ] **Surprise Treats:** A free appetizer or drink from top-rated spots once a month.

Section 5: The "Golden Question"

7. If TableGram disappeared tomorrow, how disappointed would you be?

- [ ] Very disappointed (I use it daily).
- [ ] Somewhat disappointed (I'd miss certain features).
- [ ] Not disappointed (I'd just use another app).

Survey Quantitative Analysis

Our survey simulated collection of **two main types** of data, and each type required a **different** approach.

Categorical Questions (Q1, Q3, Q4, Q6)

These are the **multiple-choice, single-select** questions (e.g., "What is the #1 reason you choose a competitor over us?"). Each respondent picked one option, producing count data.

- **Analysis:** We started with frequency counts and percentages to see the overall distribution (this is what produces charts like "42% cite cheaper delivery fees"). To go further, we use a **chi-square test of independence** to check whether response patterns

differ across customer segments. For example: do users who hold a competitor subscription (Q3) cite different switching reasons (Q4) than users who don't?

- **How it works:** Chi-square compares the observed counts in each cell of a cross-tab table against the counts we would expect if the two variables were unrelated. If the difference is large enough ($p < .05$), we conclude the variables are associated, not independent.
- **Key assumption:** Every expected cell count must be at least 5. If any cell falls below that threshold, we switch to Fisher's Exact Test, which handles small samples without approximation.
- **What this tells us:** Whether we need **different** retention strategies for different customer segments, or whether **one strategy fits all**. If subscribers switch for fundamentally different reasons than non-subscribers, a one-size-fits-all loyalty program will underperform.

Ranking Question (Q2)

Question 2 asked users to rank five app-selection drivers (cost, speed, reliability, selection, support) from most to least important. Rank data is ordinal: we know the **order** but can't assume equal **spacing between** ranks.

- **Analysis:** We reported the mean and median rank for each attribute across all respondents. This produced the **MaxDiff-style bar chart** (Total Cost = 4.5, Reliability = 4.2, etc.). To test whether the top-ranked attributes are *significantly* more important than lower-ranked ones, rather than just appearing higher due to random variation, we used a non-parametric comparison of rank distributions.
- **What this tells us:** Whether the product team should focus on ETA accuracy vs. raw speed vs. cost reduction. The chart suggested that **reliability matters more than speed**. The statistical test tells us whether that gap is real or noise.

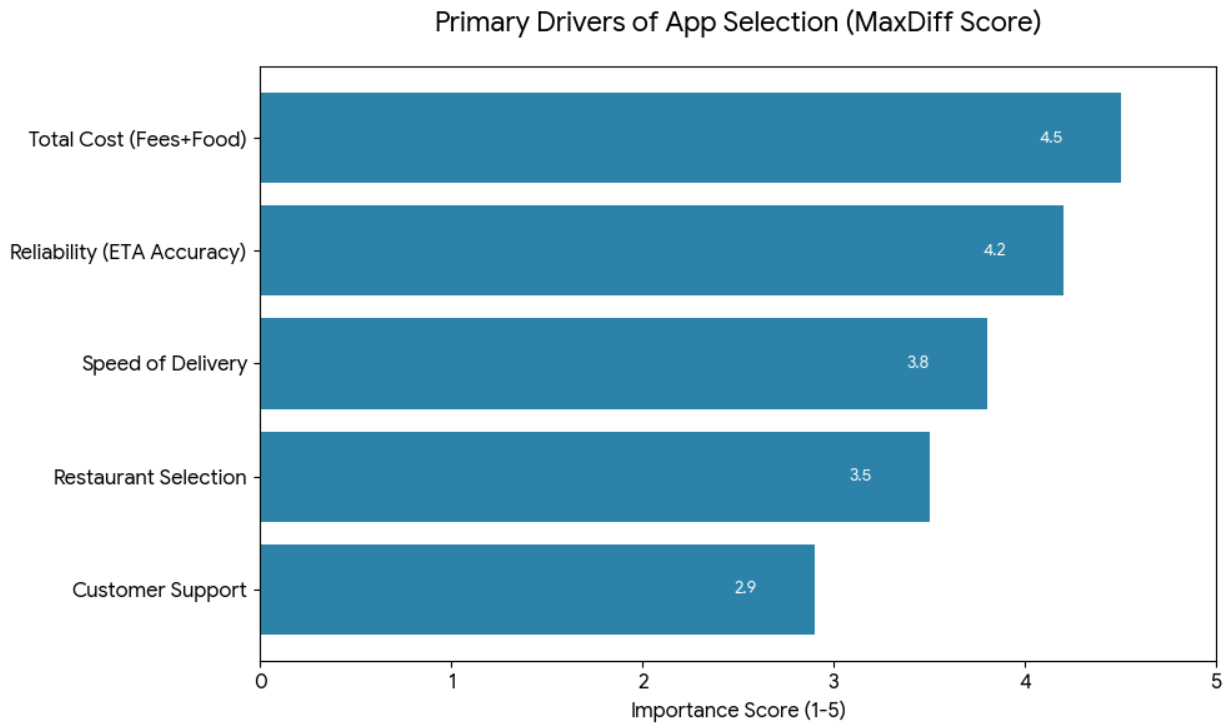
Friction Frequency (Q5) and Disappointment (Q7)

Q5 captures how often users experience **specific** problems (cold food, late delivery, hidden fees, missing items) on a Never/Once/Frequently scale. We reported the proportion of users selecting "Frequently" for each friction type and compared them. If **hidden fees** were reported as “frequent” at a significantly higher rate than other problems, that issue becomes the top-priority fix.

Q7 (the "Golden Question") produces a **three-category distribution**: Very Disappointed / Somewhat / Not Disappointed. The key insight is the overall split. Industry benchmarks suggest that **40% "Very Disappointed"** indicates strong product-market fit. Our **10% result** quantifies the gap and gives the C-suite a concrete number to act on.

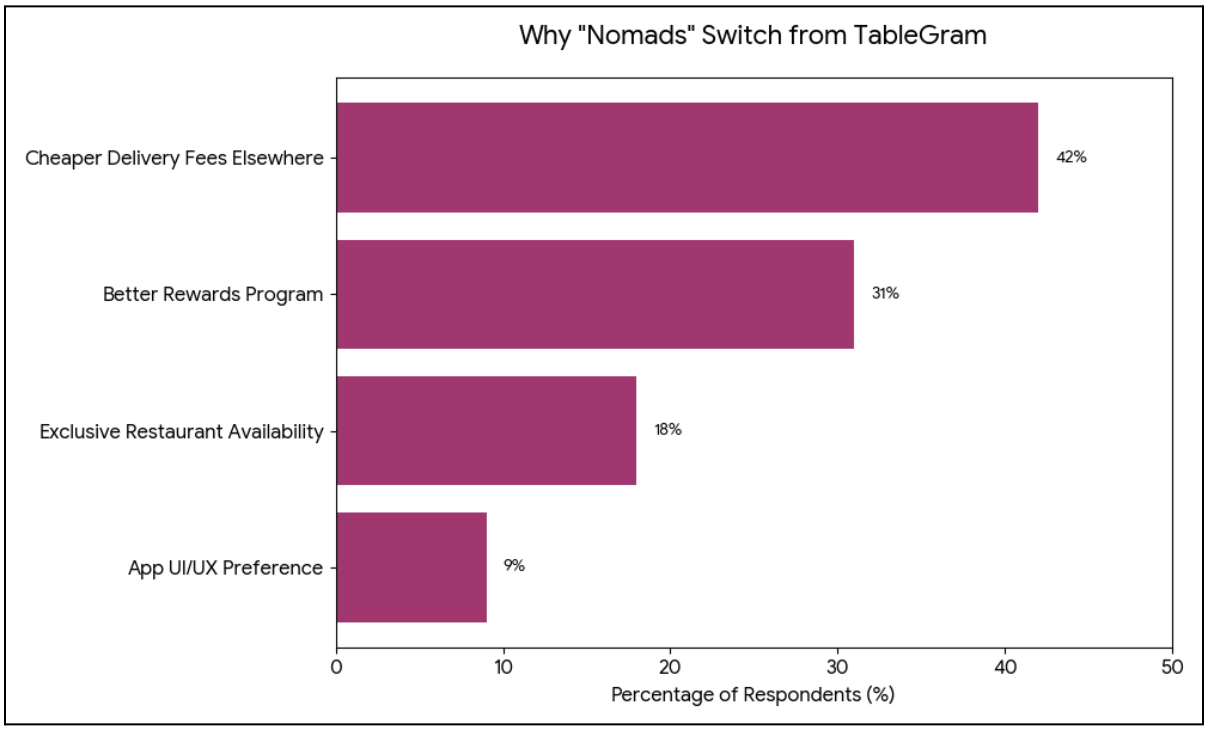
Survey Insights

Chart 1: Primary Drivers of App Selection



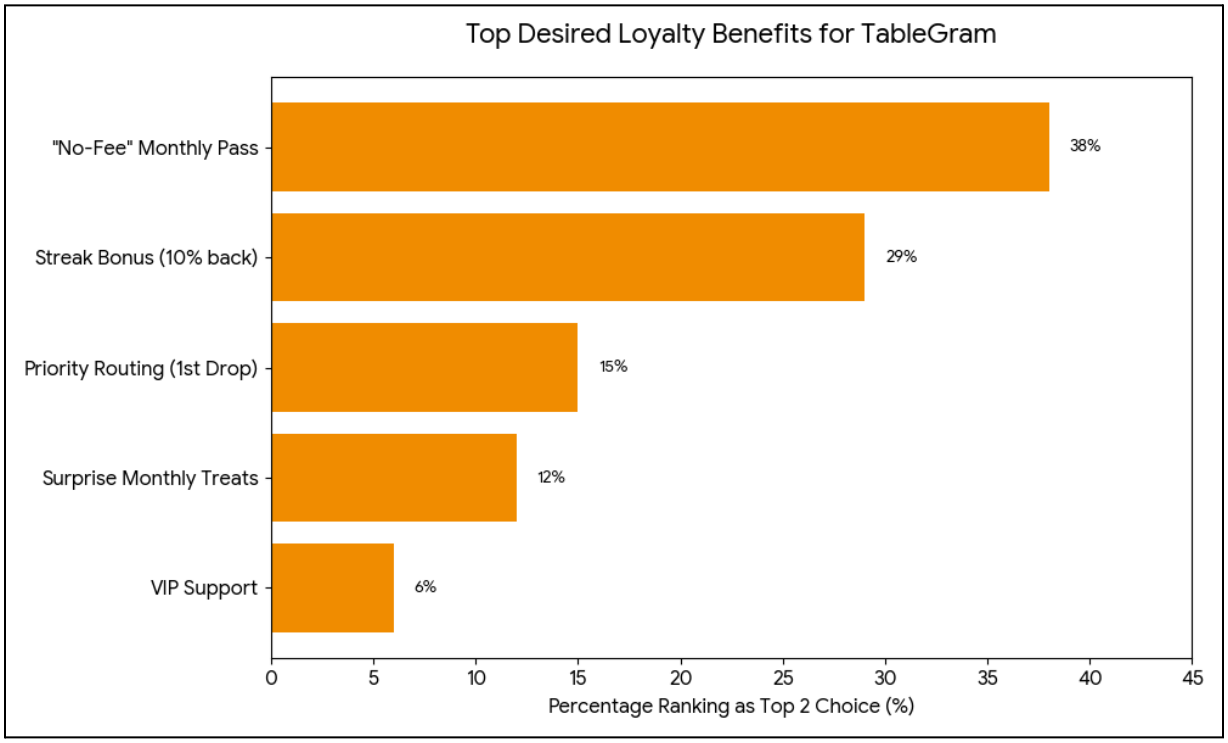
- **What it shows:** Total cost is king, but **Reliability (ETA Accuracy)** scores higher than **Speed**.
- **The C-Suite Pitch:** We don't necessarily need to be the *fastest* in the market, but we must be the most *honest*. Reliability is a driver of trust, and trust is the foundation of loyalty.

Chart 2: Why "Nomads" Switch to Competitors



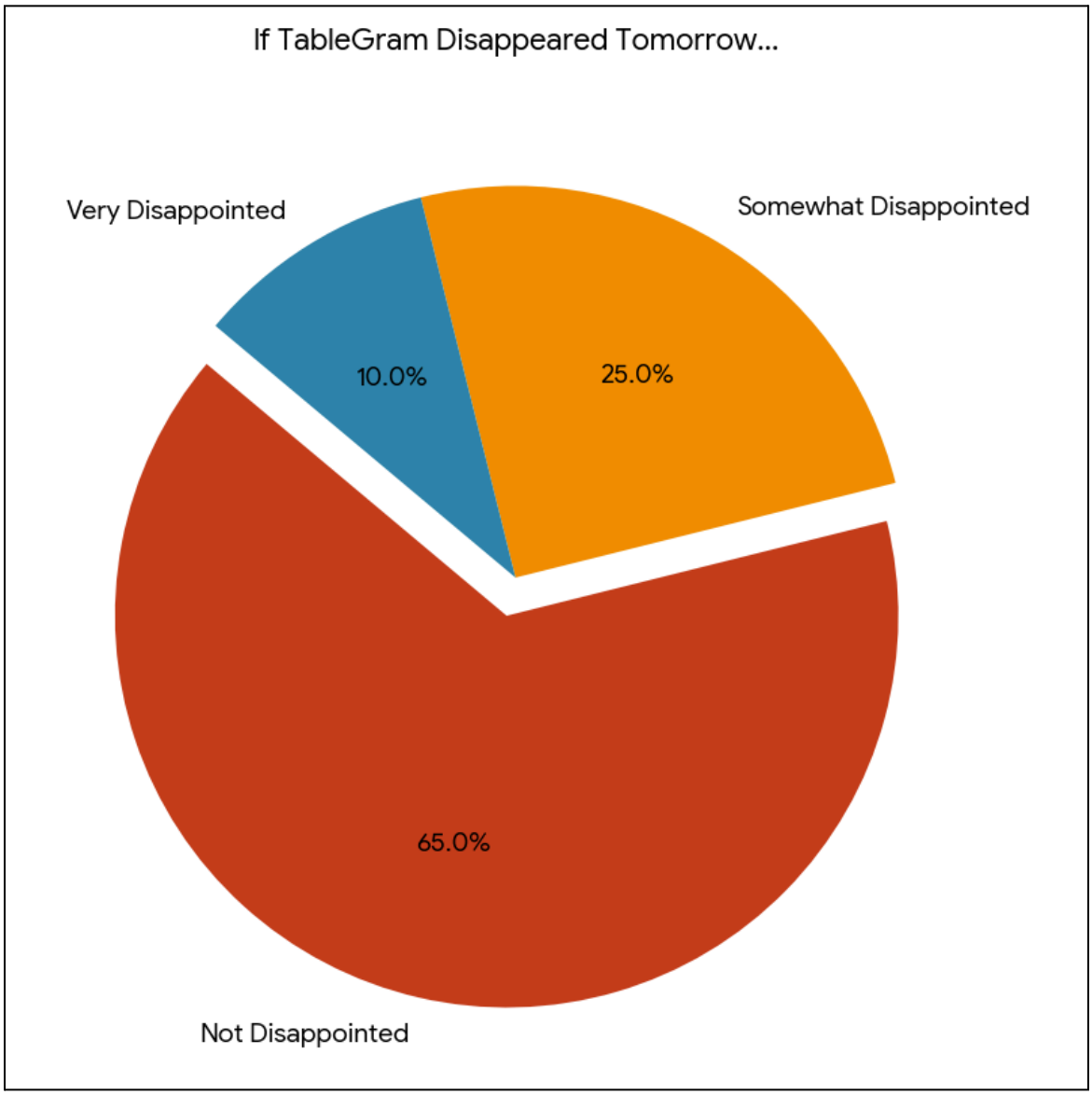
- **What it shows:** While "Cheaper Delivery Fees" is the #1 driver for switching (42%), "Better Rewards Programs" is a close second (31%).
- **The C-Suite Pitch:** We can't always win a price war on delivery fees, but we can close the 31% gap by building a **rewards ecosystem** that makes it "expensive" for a user to leave (accrued value).

Chart 3: Top Desired Loyalty Benefits



- **What it shows:** The "No-Fee Monthly Pass" (38%) and "Streak Bonuses" (29%) are the clear winners.
- **The C-Suite Pitch:** To stop the "leaky bucket," we should pivot our marketing spend. Instead of mass-distribution 20% coupons, we should invest in a **subscription model (The No-Fee Pass)** and **gamified retention (Streak Bonuses)**.

Chart 4: The "Commodity Trap" (Disappointment Metric)



- **What it shows: 65%** of our "Nomad" customers would **not** be disappointed if our service disappeared tomorrow.
- **The C-Suite Pitch:** This is our biggest risk. We are currently a **utility, not a brand**. We are winning on **price** (which is expensive for us) rather than on **relationship** or **unique value**.

Executive Strategy Summary (The "So What?")

Based on these projected findings, our recommendation to the C-Suite is to consider one of the following:

1. **Shift to Subscription:** Launch a pilot for a "No-Fee Pass." This locks the user into our ecosystem and removes the "per-order friction" that causes them to check other apps.
2. **Gamify Retention:** Implement "Streak Bonuses" (e.g., 3 orders in 14 days = 10% credit). This targets the "Nomad" behavior and rewards consistency rather than one-off discount hunting.
3. **Maximize "Reliability Score":** Focus the Product team on ETA accuracy over raw speed. If we can guarantee an arrival time more accurately than competitors, we become the "safe choice" for high-stakes meals (e.g., family dinners, office lunches).

A/B Test

Analysis of the survey results suggests that **at least one change** to the way customers interact with the **TableGram** app will improve retention and reduce churn. They also suggest that our current competitive edge is not sustainable, because it revolves around **low prices** that reduce our margins and do not offer a **qualitative benefit** over other products in the same space. Three possible interventions emerged from our analysis, as detailed in the preceding summary; we propose to implement one of them on a trial basis and assess its effect on performance through an **A/B test** with two randomly selected pools of users.

Proposal

Of the three possible interventions above, we determined that the **gamification** strategy would represent the most **lightweight** option for our developers to implement quickly and with the least impact on the app's overall performance.

Therefore, we tested a new feature that we are labeling "**Streak Bonuses**:" users who complete **3 orders within 14 days** receive **10% back** in app credit to be applied to future orders.

The test was designed as follows:

Parameter	Value
Group A (Control)	**Current** app experience, no streak mechanic. **100 users**.
Group B (Variant)	**Streak Bonus enabled**: visible progress tracker, 10% credit after 3 orders in 14 days. **100 users**.
Duration	6 weeks
Primary Metric	Order frequency (total orders per user over 6 weeks)
Assignment	Random, stratified by baseline order frequency to ensure **balanced** groups.

Hypotheses

Null Hypothesis (H_0): There is **no difference** in mean order frequency between Group A and Group B. Any observed difference is due to chance.

Alternative Hypothesis (H_1): Group B (Streak Bonus) has a **significantly higher** mean order frequency than Group A.

A/B Test Quantitative Analysis

Statistical Test: Independent Samples t-Test

We are comparing the means of two independent groups on a continuous outcome (order count). An **independent two-sample *t*-test** is the standard tool for this.

Formula (Welch's *t*-test):

$$t = \frac{\Delta \overline{X}}{s_{\Delta \bar{X}}} = \frac{\overline{X}_1 - \overline{X}_2}{\sqrt{s^2_{\bar{X}_1} + s^2_{\bar{X}_2}}},$$

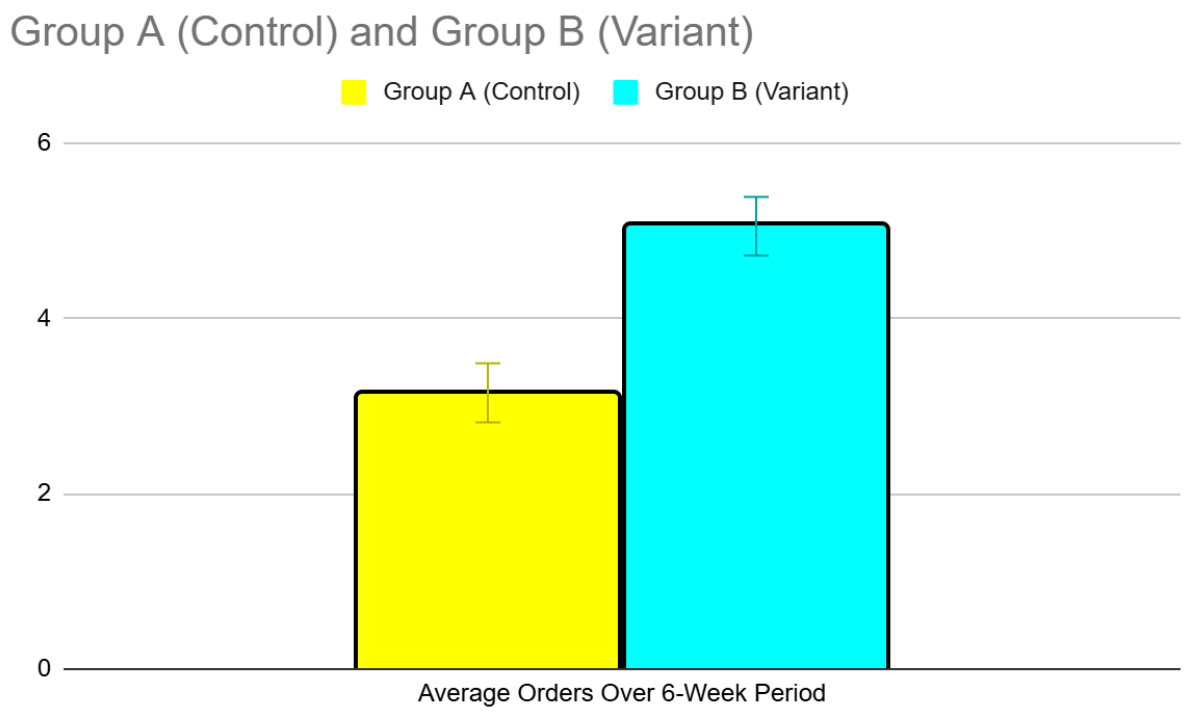
In plain terms: we divide the **difference** between the two group means by a measure of how much **natural variation** exists in the data. If the difference is **large** relative to the **noise**, the ***t*-value** is large and the ***p*-value** is small.

Assumptions We Need to Check

Assumption	What It Means	What If It Fails?
Independence	Each user's data is unrelated to other users' data. Guaranteed by random assignment at the account level.	If violated (e.g., shared accounts), the test is invalid. This is a design issue, not a statistical fix.
Normality	The order counts in each group should be roughly normally distributed. We check with a Shapiro-Wilk test or by looking at histograms.	With 100 users per group, the Central Limit Theorem makes the *t*-test robust to moderate non-normality. Only severely skewed data is a problem.
Equal Variances	Both groups should have similar spread. We check with Levene's test.	If variances are unequal, we use Welch's *t*-test (a slight modification that doesn't assume equal variances). Safe to use Welch's by default.

Results: Data & Visualization (simulated)

Full simulated **A/B test data + Analysis results** are in a spreadsheet [here]



Metric	Value
Group A Mean (Control)	**3.15** orders over 6 weeks
Group B Mean (Streak Bonus)	**5.05** orders over 6 weeks
Difference	**1.90** additional orders
p-value	**< .001** (well below the .05 threshold)
95% Confidence Interval	**[1.56, 2.24]** additional orders

Interpretation: Users with the Streak Bonus placed about **1.9 more** orders over 6 weeks than users without it. This difference is statistically significant ($p < .001$), meaning there is less than a 0.1% chance we would see a gap this large if the Streak Bonus had no real effect. The confidence interval tells us we can be 95% sure the **true effect** for our full user base falls **between 1.56 and 2.24** additional orders.

Financials: The average TableGram order generates approximately **$0.90** in average adjusted EBITDA per order, with about **$0.30** GAAP net profit per order. At an average of 1.9 additional orders per 6-week period, the Streak Bonus is projected to generate approximately $0.57 per

user over that period, just over **$0.09 per user per week and $46.80 per year** in additional profits, compared to existing baseline.

Effect Size: How Big Is This?

A *p*-value tells us the effect is real, but **doesn't** tell us how *big* the effect is. For that insight, we need to calculate Cohen's *d*:

$$d = \frac{\bar{x}_1 - \bar{x}_2}{s}.$$

d = Difference (in means) / Pooled Standard Deviation = 1.90 / 1.71 = **1.11**

Benchmarks for Cohen's *d* analysis:

d = 0.2 = small

d = 0.5 = medium

d = 0.8 = large

Our ***d* = 1.11** is a **large** effect. The Streak Bonus **shifted** ordering patterns by more than one standard deviation. This is a strong practical result that justifies a **full rollout**.

Caveat: our sample size of 100 simulated users is right around the minimum useful size for obtaining Cohen's d, *and in a real-world application we would in all likelihood strive for larger groups in an A/B test in order to obtain results as trustworthy and actionable as possible.*

Analysis: Questions and Answers

Every analysis step exists to answer a **specific business question**. The table below maps the full pipeline from method to decision.

Analysis	Data Source	What We Learn	Business Decision
Frequency counts & percentages	Survey (all Qs)	What behaviors and preferences are most common	Which loyalty barriers to address first
Chi-square test	Survey Q1 x Q3, Q3 x Q4	Whether different customer segments switch for different reasons	Do we need segmented retention strategies or one approach?
Rank comparison	Survey Q2	Which app-selection drivers matter most	Product priority: invest in ETA accuracy, speed, or cost?
Proportion comparison	Survey Q5	Which friction types are most frequent	Top operational fix (e.g., fee transparency vs. delivery speed)
Disappointment benchmark	Survey Q7	How far we are from product-market fit	Urgency level for loyalty investment (C-suite risk metric)
Independent *t*-test	A/B test	Whether Streak Bonuses increase order frequency	Go/no-go on full rollout
Cohen's *d* (effect size)	A/B test	How large the effect is in practical terms	Scale of investment: aggressive rollout vs. cautious expansion
Confidence interval	A/B test	Range of likely true effect for full user base	Revenue forecasting: expect 1.56–2.24 extra orders per user per 6 weeks

This plan ensures that every number we report connects to a decision. The survey tells us **what to fix** and **for whom**. The A/B test tells us **whether our fix works** and **by how much**.

Executive Summary & Recommendations

Key Takeaways

Problem: Low order frequency and lacklustre customer loyalty.

Solution: Introduce “Streak Bonuses”

Impact: A 60% increase in average orders with high statistical confidence ($P < 0.001$).

By adding the “Streak Bonus” feature to the user experience, we saw a **statistically significant 60% increase** in average order frequency ($p < 0.001$). We ran an A/B test on a pool of 200 users and found that our proposed change resulted in greater customer loyalty, with users moving from an average of **around 3 orders** in a 6-week period to **over 5 orders** and transforming their experience from “utility” to “habit.” This data-driven update to the app effectively addresses the primary business goal of increasing long-term customer loyalty and lifetime value.

Verdict: Implement Streak Bonus Feature

The results of our A/B test provided conclusive evidence ($p < 0.001$) that users’ average order frequency increases significantly when streak bonuses are added. With a 95% confidence interval of [1.56, 2.24], we can expect approximately **two additional orders per user every six weeks**.

Recommendation: Transition the variant (Group B) version of the app, with “streak bonus” implemented, from the test environment to a <u>full rollout</u> for 100% of the user base.

Final Considerations

While the data suggest that the “streak bonus” will improve our customer loyalty, it is essential that we **continuously monitor** the app’s performance and consider **other factors** that we may not be able to understand fully with the testing data available. Furthermore, we must also be cognizant of the possibility that **other changes** to the user experience for TableGram could produce **additional improvements** in customer loyalty and total revenue.

Therefore, we conclude with the following considerations regarding the future of the TableGram product:

Insight: Loyalty Threshold

The data show that Group B users went from making an average of ~3 orders to ~5 orders. In the food delivery space, **5+ orders** can be seen as a psychological threshold beyond which the app becomes a **habit** rather than just a **utility**.

Action: Analyze the specific pieces of the new design that led to the 4th and 5th orders. How accessible/visible is the "streak bonus feature" for new vs. returning users? How easy is it to monitor their progress? Was every member of Group B aware of it? If any were not, why was that the case?

Application: Implement these specific UI/UX elements in other areas of the app.

Variance Question

While Group B users demonstrated greater loyalty to the app, with an average of just under 2 additional orders over the 6-week trial period, the **standard deviation in this group was effectively the same** as with the control group (both around 1.7). This variance suggests that the streak bonus feature was a hit with some but not all users; perhaps it was confusing, unexpected, or not even noticed by this subset.

Action: Conduct a follow-up qualitative focus group with users who fell on the lower end of Group B (those who ordered 3 or fewer times).

Application: Refine the UI to reduce the cognitive load for the lower-frequency users in order to ensure that streak bonuses are accessible for everyone, not just regulars.

Gaming the System

Some users may place **cheap minimum orders** just to hit the streak, producing little to no improvement in total order revenues as a result of the new feature. It's even possible that users order **less value** in food overall because their focus in using the app has shifted from its **utility** as a food delivery to the **gamified usage metrics** and the desire to "win" discounts.

Action: Check whether the average order value dropped in Group B compared to Group A.

Novelty Effect

While our data indicate that introducing streak bonuses to Group B produced greater customer loyalty to the app, it is possible that this increase in usage was a function of the feature's novelty; in other words, some users may have preferred the new feature **simply because it was new.**

Action: Schedule a longitudinal study 3-6 months after rollout to determine whether the increased loyalty remains stable after streak bonuses are no longer new.

Metric: Monitor the churn rate of Group B users compared to the pre-rollout average.

Future Directions

Given that streak bonuses appear to have been successful at driving volume, we see opportunities to implement **additional** features or changes in the future.

***Personalization**: User rewards could be tailored to individual users' tastes; for instance, a demonstrated preference for Thai could earn a special discount applicable to Thai food after a certain number of orders.*

***Variety**: As an alternative, user rewards could also incentivize trying new and different cuisines; for example, users could "collect" different cuisines and receive discounts for their adventurous palate.*